INITIAL PROJECT DOCUMENTATION

1. Project Title

QuickCheck: A Student Assignment Tracking System

QuickCheck is a very simple program that helps students to organize and keep track of their assignements.. QuickCheck lets students know what assignments are due, which ones they have completed, and which ones need to be accomplished. This program was created to help students be responsible and organized in their academics.

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2. **Problem Statement**

Students often have assignments from several subjects at the same time, making it difficult to remember deadlines and track which tasks are finished. Some students rely on memory or handwritten reminders, which can result in missed deadlines, disorganized tasks, and difficulty identifying urgent assignments. This project aims to create a simple assignment tracking system that allows students to organize their tasks by subject, monitor their completion status, and identify overdue assignments.

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3. **Project Objectives**

The project aims to:

- Develop a simple system that helps students organize their assignments.
- Allow users to add and manage different subjects.
- Store important assignment information such as the subject, title, due date.
- Allow students to view assignments either as a complete list or by subject.
- Allow users to mark assignments as completed.
- Allow users to delete assignments that are no longer needed.
- Identify and display overdue assignments.
- Provide deadline warnings for urgent, upcoming, and overdue assignments.
- Recommend which assignments students should work on first based on their deadlines.
- Display the number of unfinished assignments for each subject.
- Track and display assignment completion progress for each subject.

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4. **Target Users**

Primary Target Users: Students

The system is for students who need a simple way to organize and monitor assignments from multiple subjects.

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5. **Planned Features**

1. **Add Subject** – Allows users to add subjects to the system.
2. **Add Assignment** – Allows users to record an assignment's subject, title, and due date.
3. **View All Assignments** – Displays all recorded assignments.
4. **View Assignments by Subject** – Allows users to select a subject and view its assignments.
5. **Edit Assignments** – Allows users to select a subject and edit an assignment's information.
6. **Mark Assignment as Completed** – Allows users to update an assignment's status when finished.
7. **Delete Assignment** – Allows users to remove assignments that are no longer needed.
8. **View Overdue Assignments** – Displays unfinished assignments whose deadlines have passed.
9. **Deadline Warning System** – Notifies user overdue, near deadline assignments
10. **Assignment Recommendation** – Recommends which assignment to work on first based on its deadline.
11. **Academic Workload Checker** – Counts and displays the number of unfinished assignments for each subject.
12. **Subject Progress Tracker** – Displays the number and percentage of completed assignments for each subject.
13. **Exit** – Closes the program.

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6. **Planned Inputs and Outputs**

Inputs

The user will provide:

- Subject name
- Assignment title
- Due date
- Menu choice

- Selected subject
- Selected assignment
- Updated assignment information when editing an assignment

Outputs

The program will display:

- List of all assignments
- List of available subjects
- Assignments grouped by subject
- Assignments under a selected subject
- Assignment information for editing
- Updated assignment details
- Assignment status (Completed/Not Completed)
- Overdue assignments
- Deadline warnings for urgent and upcoming assignments
- Recommended assignments to work on first
- Number of unfinished assignments for each subject
- Completion progress and percentages for each subject
- Confirmation messages

7. Logic Plan

START

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Display Main Menu

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User selects an option

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Check selected option

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Perform the chosen action

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Display result

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Return to Main Menu

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Is the choice "Exit"?

├── NO → Repeat menu

└── YES → END